

Machine Learning Quiz Guide

1. Linear Regression

Definition: Linear Regression is a supervised machine learning algorithm used to predict a continuous numerical value using a linear relationship between input and output variables.

Explanation: It fits a straight line through data points and predicts numeric values such as salary, marks, or house prices.

Formula: $y = mx + b$

Study Hours	Marks
1	40
2	50
3	60

Example equation: $y = 10x + 30$ If $x = 4$ hours, predicted marks = 70.

2. Logistic Regression

Definition: Logistic Regression is a supervised machine learning classification algorithm used to predict categories or classes.

Explanation: It predicts probabilities between 0 and 1 and is commonly used for spam detection, disease prediction, and pass/fail problems.

Formula: $\sigma(z) = 1 / (1 + e^{-z})$

Email Text	Spam
Win money now	1
Meeting at 5 PM	0

If the model predicts probability = 0.85, then the email is classified as Spam.

3. Clustering (K-Means)

Definition: Clustering is an unsupervised machine learning technique used to group similar data points together without labeled data.

Explanation: K-Means clustering groups data into clusters using distance calculations.

Formula: $d = \sqrt{(x - x_c)^2 + (y - y_c)^2}$

Customer	Spending
A	Low
B	Low
C	High
D	High

K-Means may create two clusters: low spenders and high spenders.

Final Comparison Table

Topic	Linear Regression	Logistic Regression	Clustering
Learning Type	Supervised	Supervised	Unsupervised
Output	Continuous Value	Category/Class	Groups
Example	House Price	Spam Detection	Customer Groups